

Macro Three Replication Project: Investigating the Economic Impact of the CFA Franc Zone with Synthetic Control Methods

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1 Replication Target

In this project, we aim to replicate the findings of *From the French franc to the euro, is there an economic impact for the CFA franc zone countries? A 'bias-corrected' synthetic control method* by Alassane Diallo and Adama Ba, published in Cogent Economics & Finance in 2024. This paper assesses the impact of pegging the CFA franc to the euro on real GDP per capita in the West African Economic and Monetary Union (WAEMU) and the Central African Economic and Monetary Community (CAEMC) countries. The original study uses a set of 35 annual country-level panel data from 1980 to 2019 and applies synthetic control methods to estimate the counterfactual economic outcomes for member countries.

2 Data

Table 1: Variables

Variable	Symbol	Type	Category	(First) Page Ref	Poss
Real GDP per capita	Y_{it}	Continuous	Outcome	6	
Lagged GDP per capita	$Y_{i,t-k}$	Continuous	Lagged outcome	10	
Gross Fixed Capital Formation (% GDP)	I_{it}	Continuous	Capital	10	
Government expenditure (% GDP)	G_{it}	Continuous	Fiscal	10	
Official Development Assistance (% GDP)	ODA_{it}	Continuous	External	10	
Foreign Development Aid (% GDP)	FDA_{it}	Continuous	External	10	
Employment rate	L_{it}	Continuous	Labour	10	
Agriculture (% GDP)	$Agri_{it}$	Continuous	Structure	10	WDI
Industry (% GDP)	Ind_{it}	Continuous	Structure	10	WDI
Governance / institutional quality	$Inst_{it}$	Index	Institutions	10	
Political regime characteristics	Pol_{it}	Categorical/Index	Institutions	10	
Inflation	π_{it}	Continuous	Macro controls	5	
Euro peg dummy	D_{it}	Binary	Treatment		
Synthetic GDP per capita	$\tilde{Y}_{it}$	Continuous	Constructed		
Treatment effect	τ_{it}	Continuous	Constructed		